

Convert casual riders to annual members

Spotting differences between casual riders and annual members

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Business Task

Annual members generate higher revenue to the Cyclistic company than casual riders. Therefore, it is advisable to target casual riders with a campaign to convince them to upgrade to annual membership. The goal of the current analysis is to compare the behavior of casual riders and annual members and look for trends and differences. The insights can be used to set up a campaign that targets casual members to upgrade to annual membership. For this goal, the past 12 months will be analyzed.

Data source

Data is owned by Lyft Bikes and Scooters, LLC, and the City of Chicago. Original data can be accessed from [here](#), under the following [license](#). Datasets contain the rides of specific months. They can be downloaded as .csv files. I used the past 12 months (May 2025 – April 2026) for the analysis.

Data cleaning

Checking the columns

ride_id is the primary key; it should be unique. I checked whether there are duplicates, but I didn't find any. *rideable_type* and *member_casual* are texts; the former can be classic_bike or electric_bike, and the latter can be member or casual.

start_station_name, *start_station_id*, *end_station_name*, and *end_station_id* identify the starting station (the first two) and the end station (the last two). They are texts. Note that these columns may have missing values. Whenever a starting station is missing, its ID is also missing. Likewise end stations. Note also that there was a change in station IDs in June 2025. From June 2025, the form of station IDs is CHIxxxxx (where xxxxx is a 5-digit number). Before that, however, station IDs were different.

start_lat, *start_lng*, *end_lat*, and *end_lng* identify GPS coordinates for starting points (the first two) and end points (the last two). They are numbers. Note that in some cases, endpoint GPS coordinates are missing (starting points' GPS is never missing). Whenever *end_lat* misses, so does *end_lng*.

The *started_at* and *ended_at* columns are date-time types of departing (*started_at*) and arriving (*ended_at*) dates and times. During the import, Excel didn't handle these columns well. So, I needed to import them as texts and transform them into proper date-time format. Their format is yyyy-mm-dd hh:mm:ss.000.

With that, every column contains the correct data and type.

Remove data where end point GPS is missing

Missing endpoint GPS can be problematic, so I removed observations when these values were missing. I should note that the biggest amount of removed data was 988 (out of 678904), or 0.15% in June 2025. It shows that removing these data won't affect the analysis noticeably.

There are many more cases when the starting station and/or end station is missing. Removing them would cause bigger data loss, so I decided to include them in the analysis.

Calculating travel time

I calculated travel time in minutes by subtracting *the started_at* value from *the ended_at* value and multiplying by 1440 (1 day). Note that Excel stores date-time as a number; the difference of two date-time values gives the difference in days. Multiplying with the correct number (1440 in the case of minutes) will give the appropriate time difference. I named the new column *travel_time_min*.

Determine starting day

I used the `=WEEKDAY(started_at, 2)` function to determine the starting day of the ride. This function returns numbers where 1 is Monday, 2 is Tuesday, ..., and 7 is Sunday. I named the new column *weekday_start*.

Removing „invalid” rides

In some cases, rides take very little time, even just some seconds, while the starting point and end point are the same. These are clearly biases. To determine which ride is valid and which isn't, there should be more data collected. Because it is not available right now, I decided I will consider those rides that didn't take at least 5 minutes invalid. Note that most probably some valid rides will be considered invalid falsely, and there can be biases even if a ride took more than 5 minutes. Nonetheless, the ratios of biases will be lower in this case.

The number of removed data points is between 42344 (December, 2025) and 151961 (November, 2025). In percentages, the best ratio of removed data is 18.44% (June 2025), and the worst is 32.39% (January 2026). The latter means that at least two-thirds of the data was always analyzed.

With these steps, I consider the data clean.

Analysis steps

I made an analysis according to 5 viewpoints. They are:

1. **Months:** I simply compared the number of rides for total rides, casual riders, and annual members according to different months.
2. **Rideable types:** I checked whether classic bikes or electric bikes are more popular among all rides, casual riders, and annual members.
3. **Type of days:** I checked which days of the week are the most popular among all rides, casual riders, and annual members.
4. **Travel time:** I checked how long the travel time is among all rides, casual riders, and annual members.
5. **Starting station:** I checked which stations are the most popular among all rides.

For the analysis, I combined BigQuery with Excel. I queried the data with BigQuery, copied the results to Excel, and modified it further.

1. Months

There are two goals of this analysis:

1. Determine which months are the most popular. A campaign should be held in those months when it can reach the most people.
2. Determine the casual riders / annual members ratios in different months. Part of the campaign can emphasize those months where the ratio is high.

With one query / dataset, I determined the number of rides for total rides, casual riders and annual members. Also, I calculated the ratios of casual riders / total rides and annual members / total rides, in percentages. I arranged those data in Excel.

2. Rideable types

This analysis also have two goals:

1. Determine whether classic bikes or electric bikes are more popular among all rides. The more popular should be preferred in a campaign.
2. Determine whether there is a difference in the preferred bike type between casual riders and annual members. If casual riders prefer a different bike type than annual members, that bike type should be in the focus of a campaign.

With two queries / dataset, the number of casual riders and annual members, as well as their percentages of the total rides, were determined when they used classic bikes (query 1) or electric bikes (query 2). Also, the ratio of classic bikes / total rides (query 1) and electric bikes / total rides (query 2) was determined. The results were copied to and arranged in Excel.

3. Type of days

There are two goals in this analysis:

1. Determine which days are most popular among all rides. Those days should be in focus.
2. Determine whether there is a difference of preference of days between casual riders and annual members. If casual riders prefer different days than annual members, those days should be in focus.

With 3 queries / dataset, the number of rides was determined for every day type in the case of casual riders (query 1), annual members (query 2), and total rides (query 3). The results were copied to and arranged in Excel. For total rides, the percentages of given days / all rides were calculated. For casual riders and annual members, the percentages of casual riders / total rides and annual members / total rides were calculated for each day type.

4. Travel time

There are two goals in this analysis as well:

1. Determine the average travel time among all rides, casual riders, and annual members. With this, it can be quickly compared which group prefers shorter and which prefers longer rides.
2. Determine travel time categories for total rides, casual riders, and annual members. With this, it can be inspected how long travels the different groups prefer. The travel time should be compared between casual riders and annual members.

With 1 query / dataset, I determined the average travel time for all rides, casual riders, and annual members. With 6 queries / dataset, I calculated the number of rides for total rides, casual riders, and annual members in different travel time categories. I used the following categories: less than 6 minutes, 6 – 10 minutes, 10 – 20 minutes, 20 – 30 minutes, 30 – 60 minutes, and more than 60 minutes. I copied and arranged the data in Excel. The casual riders / total rides and annual members / total rides were calculated for every month and every time category in percentages. The total rides for every time category / all rides were also calculated in percentages.

5. Starting station

The goal of this analysis was to determine which stations are most popular among all rides.

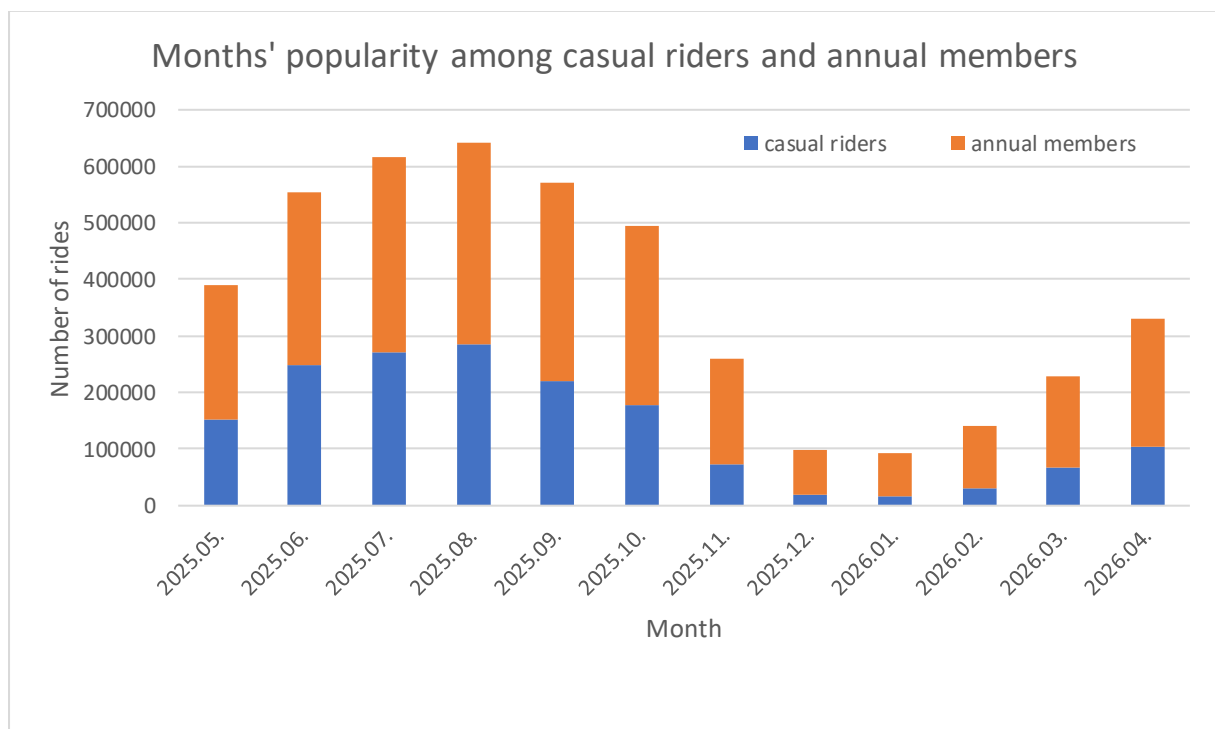
With 1 query / dataset, the total rides by starting stations were calculated in cases where the starting station wasn't missing, and the ratios of these calculated total rides per station / all rides were determined. The data was copied to and organized in Excel. The maximum percentage of the most popular station rides / total rides was collected. It was determined how many of the most popular starting stations need to be considered to cover 50% and 90% of total rides. The top 10 starting stations were collected for every month. Stations that were in the top 10 at least in 6 months were collected.

Results

For the visualizations, I used Excel.

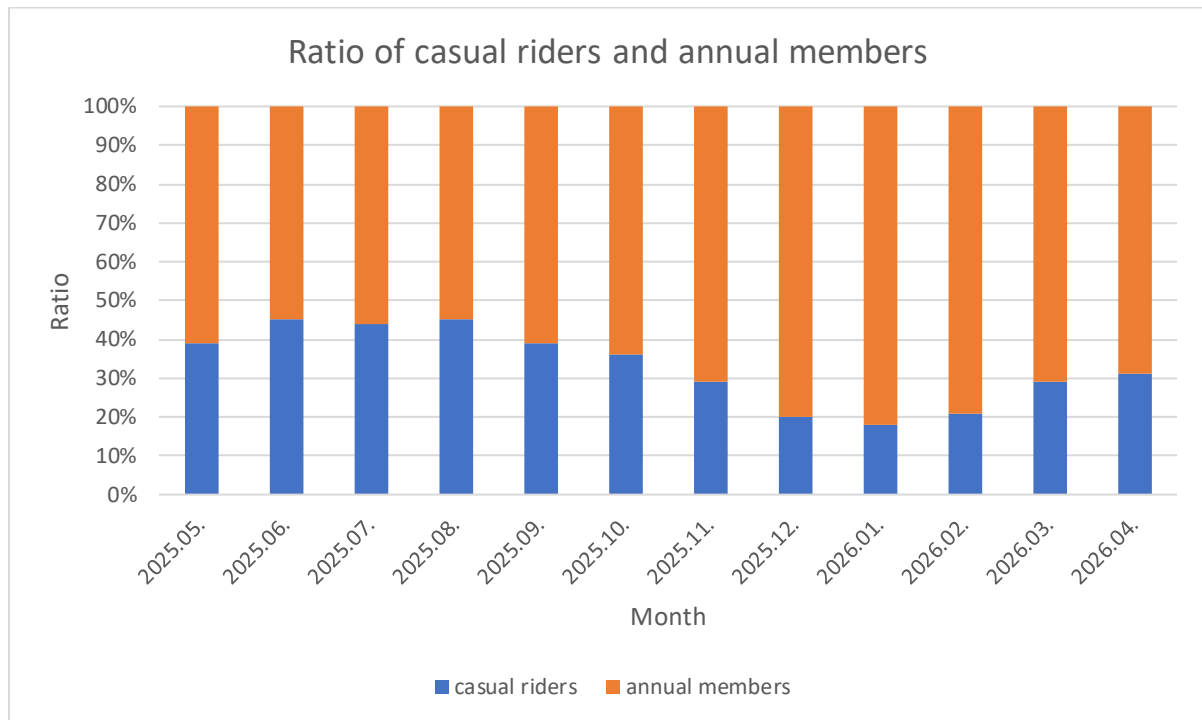
1. Months

The least popular months belong to winter, from December to February; the total number of rides is below 200,000. The most popular season is summer, from June to September; the total number of rides is above 500,000. Popularity of spring and late autumn is between the winter and summer seasons. The seasons' popularity is similar between casual riders and annual members. The number of rides vs. months can be seen on Picture 1.



Picture 1: Number of rides vs. month

The summer season is also a good target considering the ratio of casual riders and annual members. From June to August, more than 40% of the rides were done by casual riders. On the other hand, from December to February, barely 20% of the rides were done by them. Other months' ratios are between 20% and 40%. It can be seen in Picture 2.

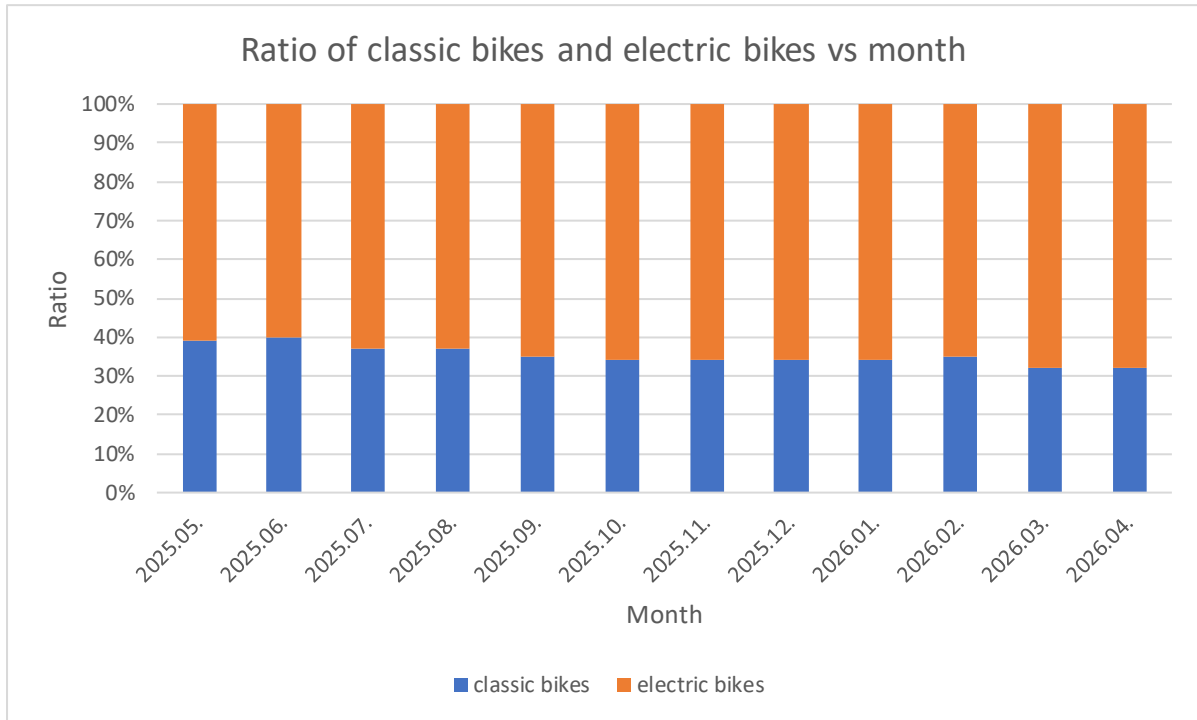


Picture 2: Ratio of casual riders and annual members vs month

Considering the total rides, summer, especially from June to September, is the most popular season. Again, the ratio of casual riders is the highest during summer, especially between June and August.

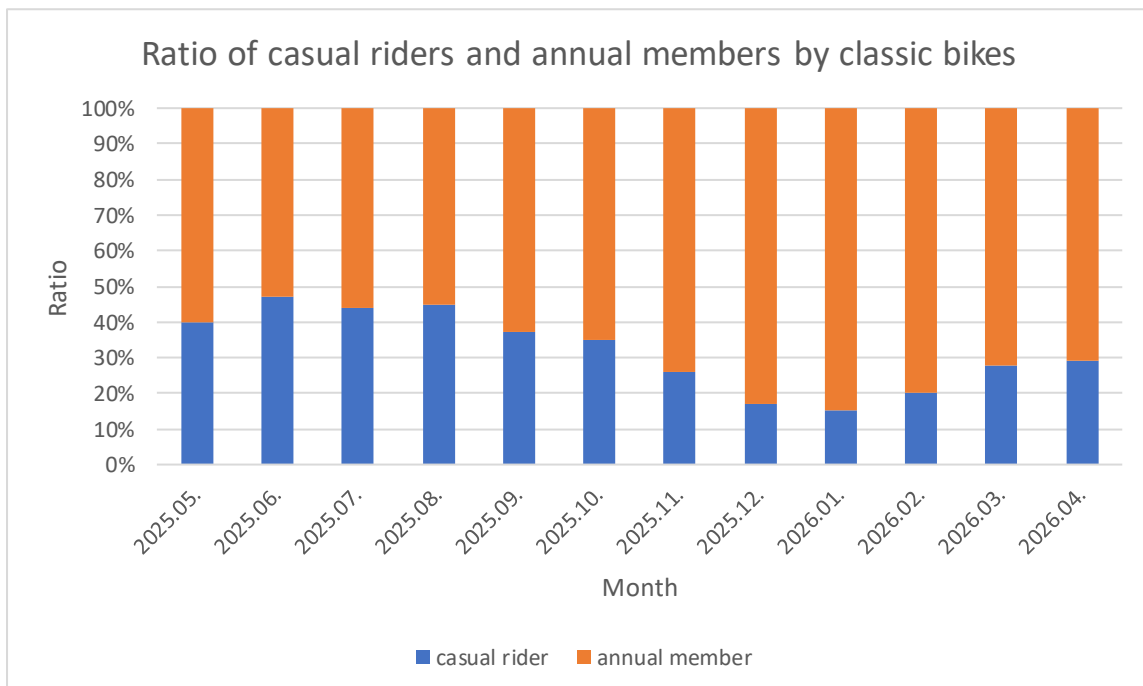
2. Rideable types

Considering the total rides, electric bikes are more popular. The ratio of classic bikes / total rides is around 30-40% (Picture 3). Also, it seems the popularity of electric bikes doesn't depend much on the month.

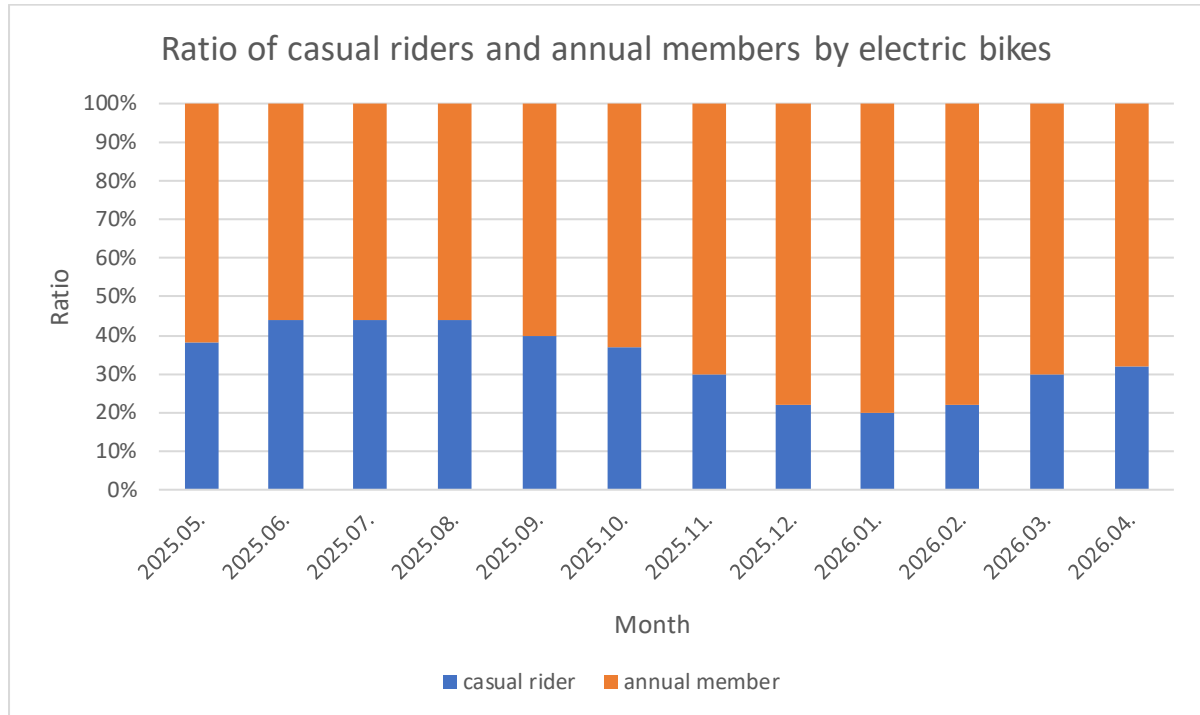


Picture 3: Ratio of classic bikes and electric bikes vs month

It seems there is no difference in bike type preference between casual riders and annual members (Pictures 4 and 5). The ratio of casual riders and annual members is nearly the same for classic bikes, electric bikes, and total rides.



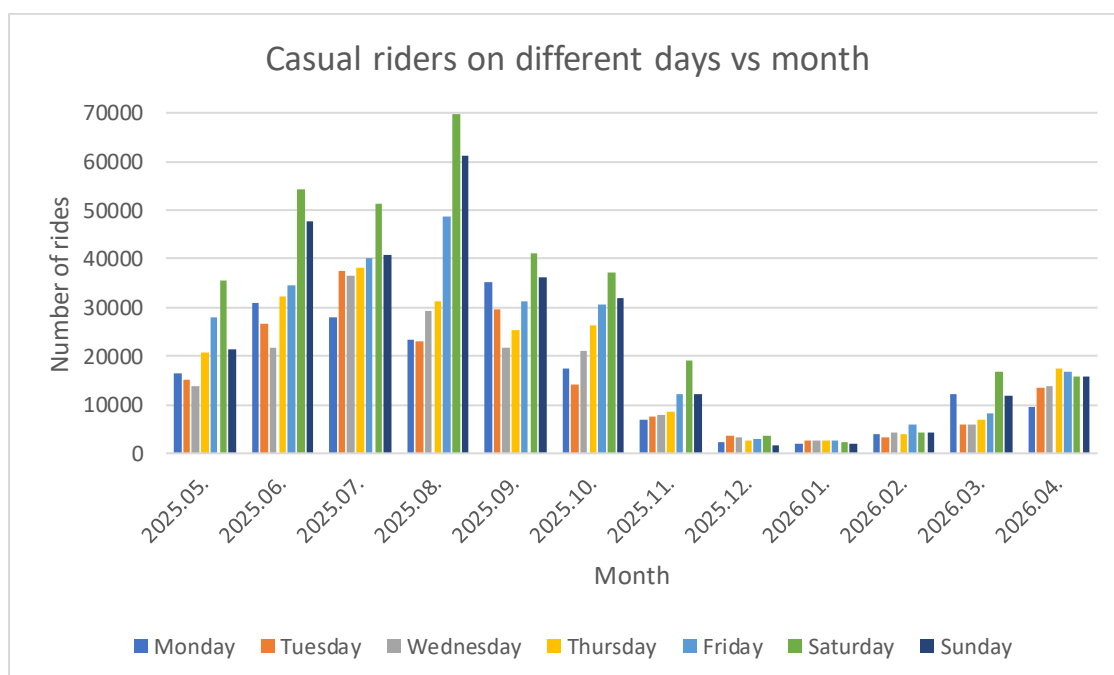
Picture 4: Ratio of casual riders and annual members vs month by classic bikes



Picture 5: Ratio of casual riders and annual members vs month by electric bikes

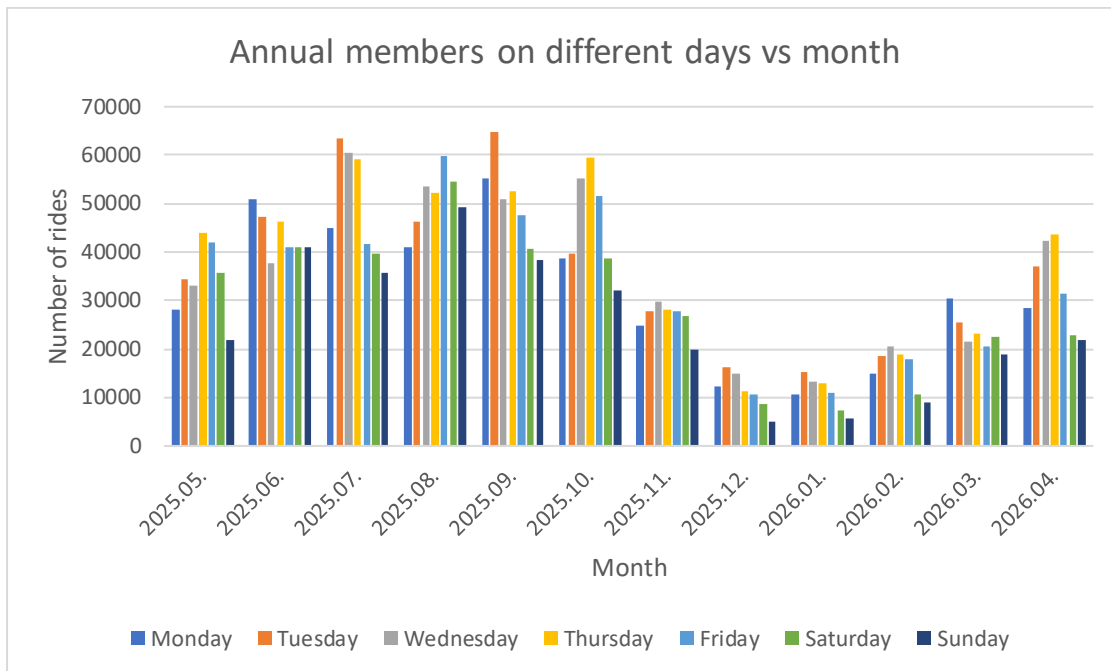
3. Type of days

Among casual riders, weekend rides (Saturday and Sunday) are the most popular, especially in the summer season. Sometimes, Friday and Monday are also more popular than the rest of the days. (See Picture 6).



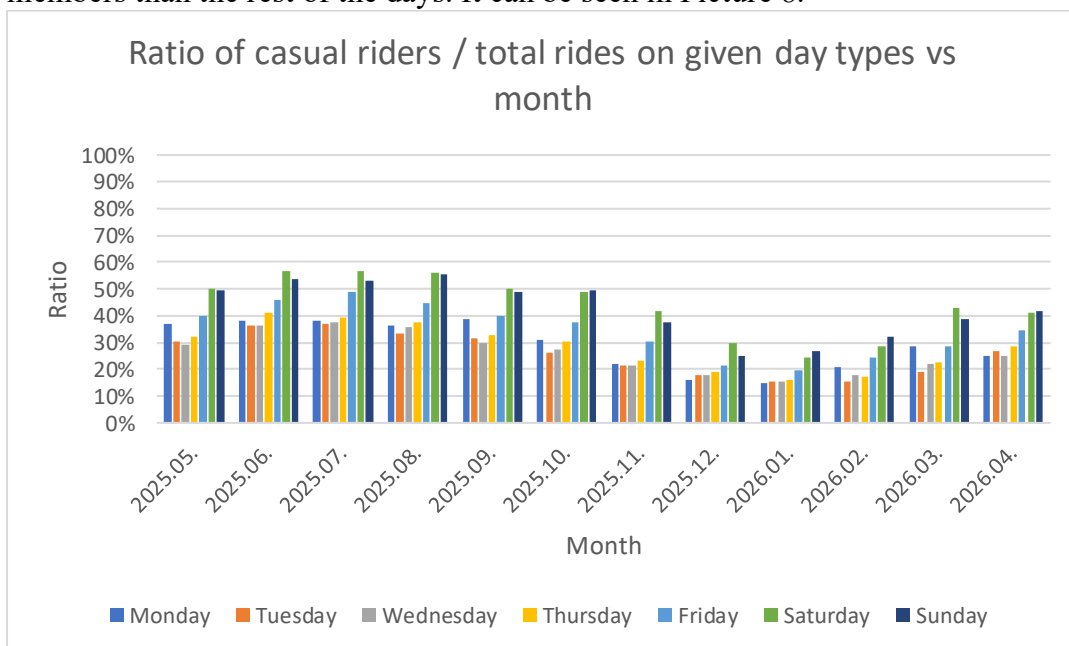
Picture 6: Number of rides on different days among casual riders

Among annual members, however, weekdays (Monday – Friday) are more popular. It can vary which day is the most popular from month to month; in some cases, some weekend days are more popular than some weekdays, but in every month, one of the weekdays is the most popular. It can be seen in Picture 7.



Picture 7: Number of rides on different days among annual members

Speaking of ratios, weekend rides are more popular among casual riders. These days have a bigger casual rider / annual member ratio in every month. It is especially true for summer season: from June to August, more than 50% of rides on Saturday and Sunday were taken by casual riders. Friday and sometimes Monday also have a bigger ratio of casual riders / annual members than the rest of the days. It can be seen in Picture 8.

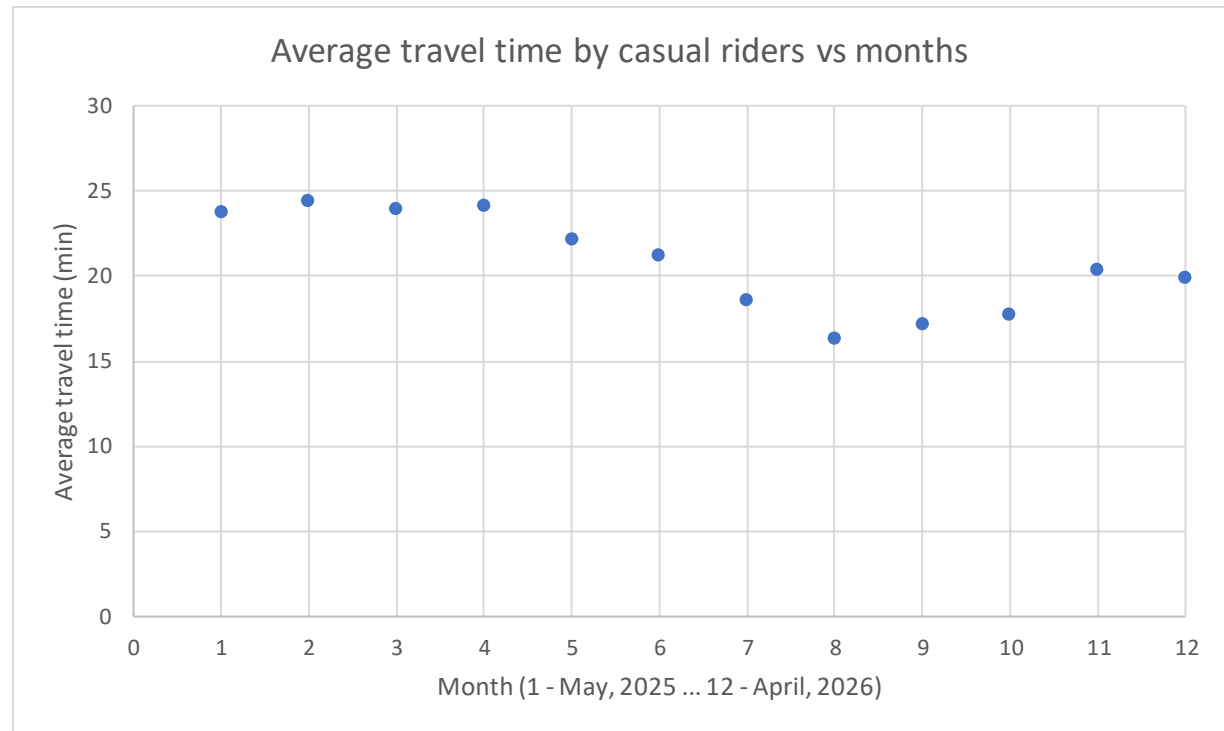


Picture 8: Ratio of casual riders / total rides on different days vs month.

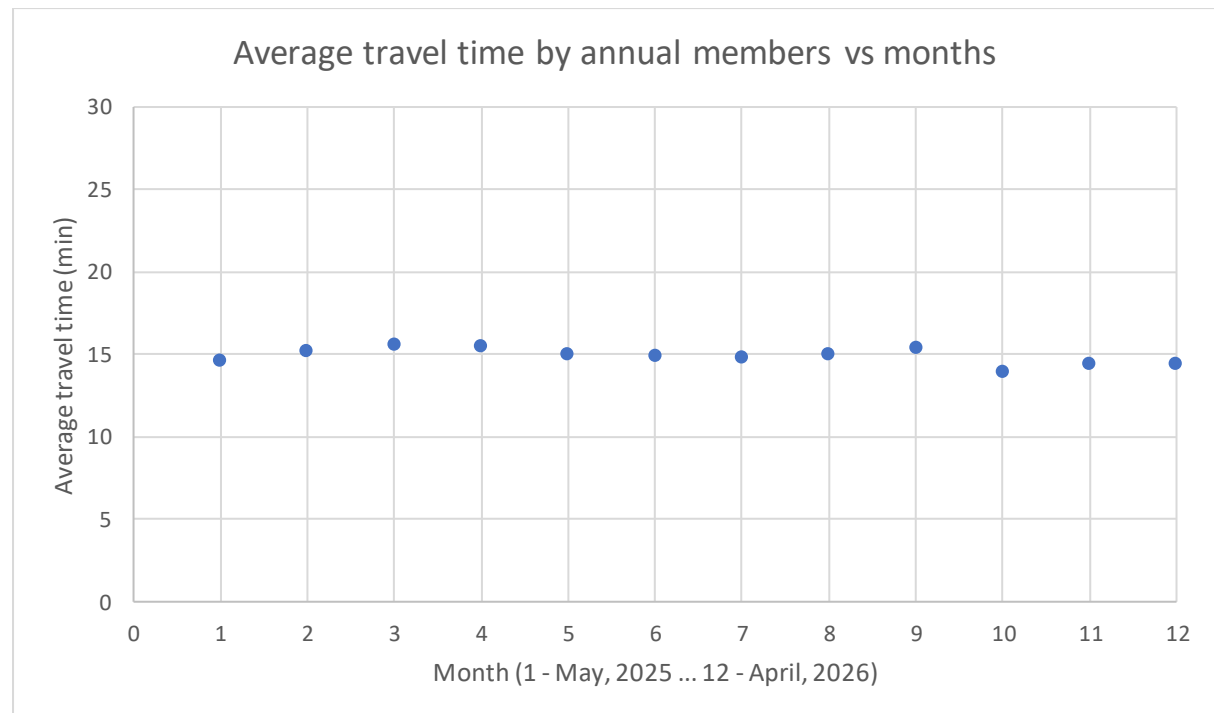
It also means that annual members have a higher ratio on weekdays, especially from Tuesday to Thursday.

4. Travel time

Casual riders tend to take longer rides. Their average travel time is between 15 and 25 minutes, depending on the month. Annual members, however, usually took shorter rides. Their average travel time is around 15 minutes through the whole year. These can be seen in Picture 9 and 10.

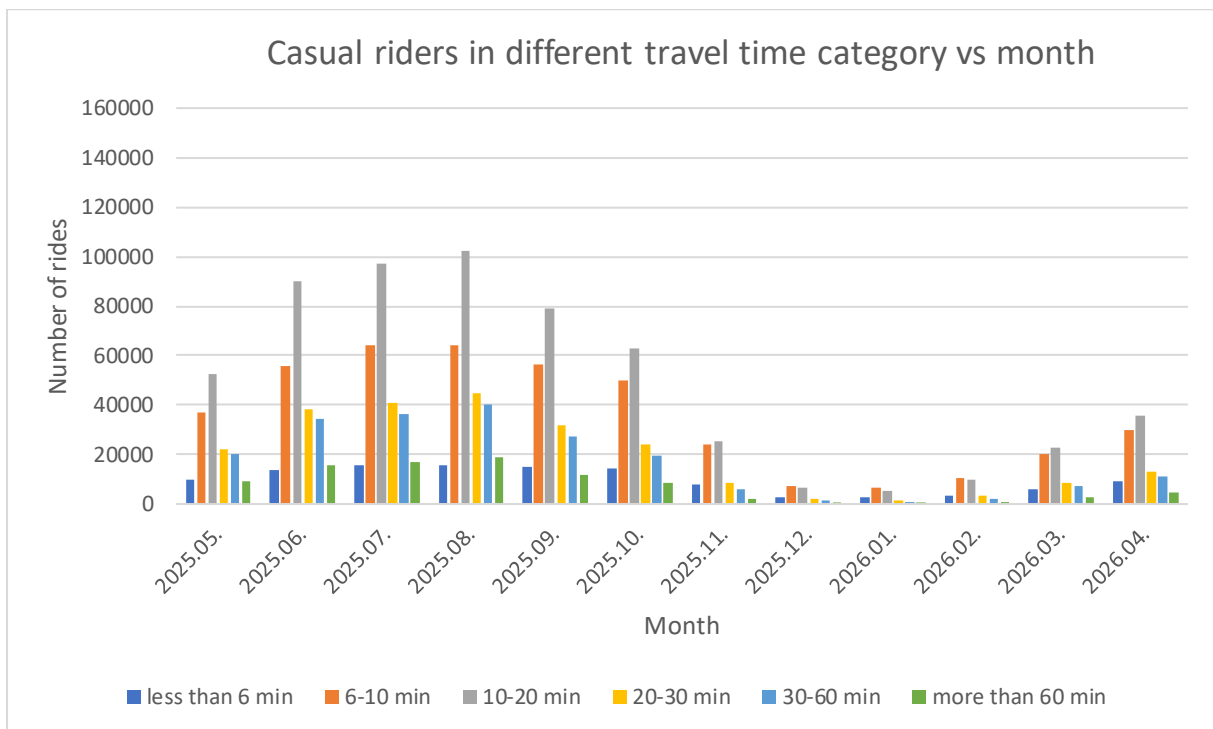


Picture 9: Average travel time by casual riders

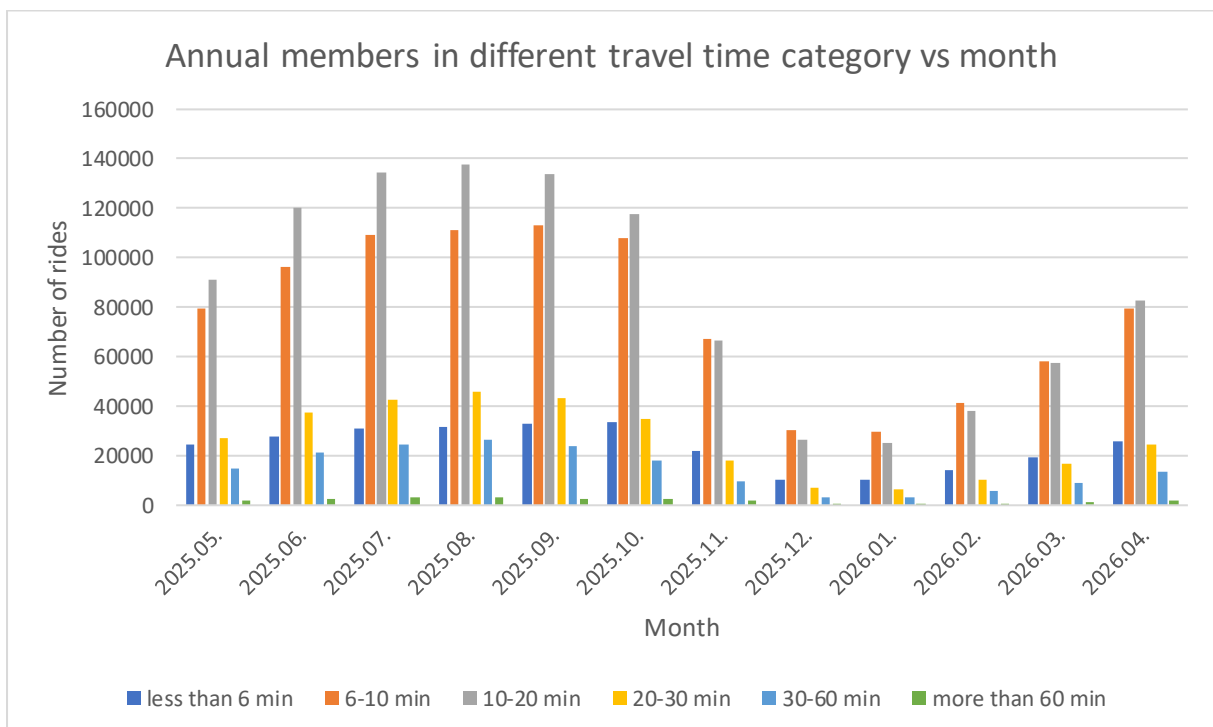


Picture 10: Average travel time by annual members

Among both casual riders and annual members, the 6 – 10 min and 10 – 20 min time frames are the most popular, especially the latter. (Pictures 11 and 12). However, the ratios differ: among annual members, the popularity of these categories compared to others is much higher than in the case of casual riders. (Reminder: fewer than 5 minutes' rides were removed during data cleaning. So, less than 6 min is actually 5 – 6 min.)

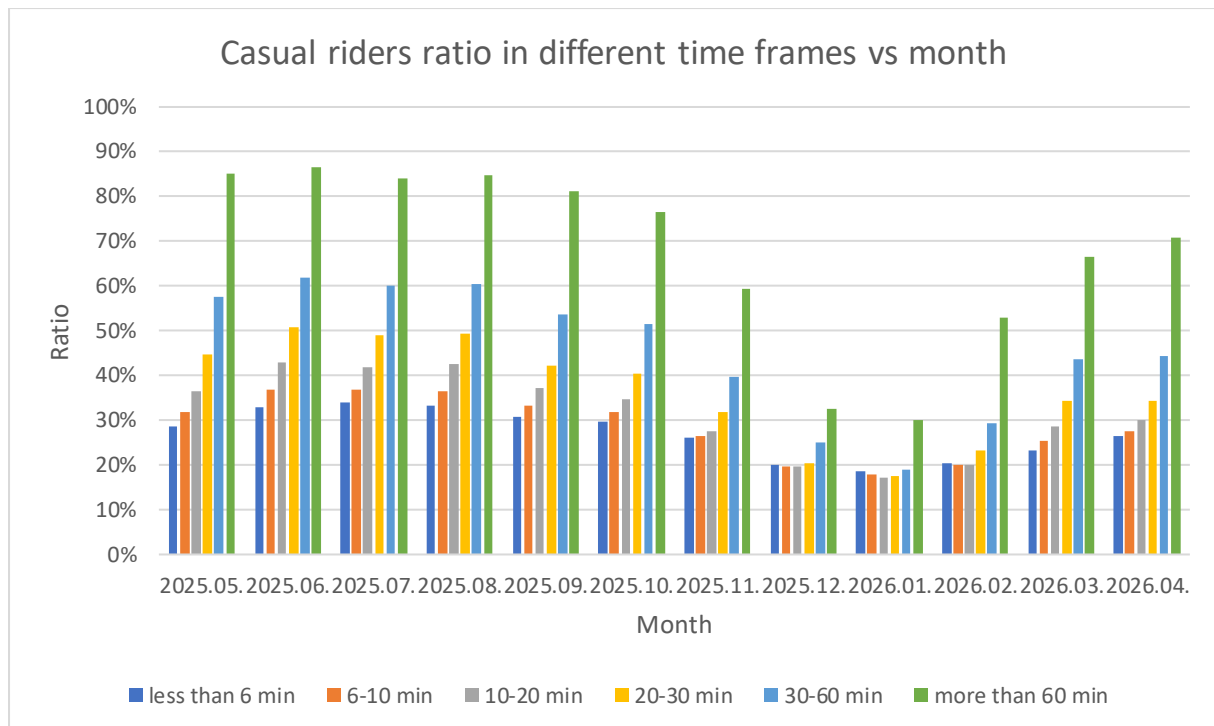


Picture 11: Number of rides among casual riders with different travel time categories



Picture 12: Number of rides among casual riders with different travel time categories

Considering the ratios, the longer the ride, the higher the ratio of casual members (Picture 13). During the winter season, it may show a slightly different pattern.



Picture 13: Casual riders ratio in different travel time categories

5. Starting station

There are over 1000 starting stations in every month, including the least popular winter seasons. Comparing them individually didn't provide useful results. In the end, I queried only total rides because it seemed further analysis (split to casual riders and annual members) would be meaningless only with these data.

The most popular station merely has around 1-2% of all rides. The highest percentage was 2.53% in August, 2025. At least the 100 most popular stations need to be considered to get 50% of all rides; the lowest value was 104 in June 2025 and August 2025, and the highest was 143 in April 2026. To get 90% of all rides, between 350 and 450 stations need to be considered; the lowest value was 351 in June 2025, and the highest was 437 in April 2026.

The 8 most popular stations, which appeared in the top 10 at least in 6 months, were *DuSable Lake Shore Dr & Monroe St*, *Michigan Ave & Oak St*, *Millennium Park*, *Canal St & Madison St*, *DuSable Lake Shore Dr & North Blvd*, *Kingsbury St & Kinzie St*, *Navy Pier*, and *Clinton St & Madison St*.

Take away

Summer is the most popular season, especially from June to September, and the casual riders' ratio is the highest in summer, too, especially from June to August.

There is no noticeable difference in the preferred bike type (classic or electric) between casual riders and annual members. However, electric bikes are more popular in common; among 60%–70% of the rides, they are electric bike rides.

Considering both the total rides and the ratios, weekends (Saturday and Sunday) are more popular among casual riders, while weekdays (Monday – Friday) are more popular among annual members.

Among both the casual riders and annual members, the 6 – 10 min and 10 – 20 min travel time categories are the most popular. However, casual riders tend to attend longer rides more willingly than annual members.

The most popular starting stations cover 1-2% of all rides. To cover at least 90% of all rides, 350 – 450 starting stations needed to be considered out of more than 1000.

Recommendations

1. A campaign should be held during summer, especially from June to August, to make annual membership more popular. In this period, it can reach the most people and have the best chance to convert casual riders to annual members, whose ratio is between 40% and 50%.
2. It is worthy to consider introducing "summer membership." It would be paid annually, just like annual membership, but it would have a slightly higher price than half of the price of the annual membership and would be valid from May to October. Casual riders probably consider annual membership a waste, as they don't use it nearly half of the year, especially in the winter season.
Before introducing it, though, a survey should be held among casual riders and annual members: whether casual riders would willingly buy summer membership and whether annual members would downgrade to summer membership. The first would increase the revenue, while the latter would decrease it. If more casual riders would upgrade than annual members downgrade, it would be a success.
3. A perk should be introduced to annual membership that targets the weekend. The weekend is more popular among casual riders, and it would be easier to convert them into annual members if they could utilize some advantages when they take a ride on the weekend.

4. A further analysis may be useful considering the starting stations. These stations should be grouped into community areas of Chicago, and we should analyze whether some community areas are more popular than others. If so, the campaign could be located in those areas and advertise annual membership there.